

Evolving Propositions: From Continuous Functions to Boolean Correspondence and Logical Matrices

Exact Abstraction, Quasianalytic Rigidity, and Operator Switching

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Abstract

We study when continuous real-valued functions induce exact propositional operators and how the resulting Boolean functions may be represented by correspondence matrices (CMs) and formula-valued logical matrices (LMs). A collection of thresholded input propositions determines an exact Boolean output operator precisely when the output is constant on every continuous state-space cell identified by the input abstraction. We formulate this as an exact-factorization criterion and specialize it to sign abstractions of real variables.

For continuous maps on nonzero orthants, connectedness gives a canonical induced Boolean function whenever the output avoids the switching boundary, and this construction respects composition. We then study the converse realizability problem. Every finite Boolean function admits an exact continuous piecewise-linear realization and an exact C^∞ realization on the open orthants. More sharply, a regularity dichotomy appears in Denjoy–Carleman scales: standard non-quasianalytic classes admit exact realizations of every Boolean function, while quasianalytic classes with coordinate-division closure can induce only affine-parity sign laws under the global nonvanishing hypothesis. Real-analytic and polynomial rigidity follow as special cases. Thus the key boundary is not smooth versus analytic as such, but the availability versus exclusion of flat functions.

For continuous functions that are not globally sign-stable on each orthant, we introduce a continuous polarity score tensor and its thresholded correspondence tensor. Magnitude or parameter space decomposes into operator chambers on which the induced Boolean function is constant. Crossing a generic switching surface changes individual truth-table entries, yielding piecewise-constant trajectories through the finite space of Boolean operators. Each exact chamber-level Boolean function canonically determines a correspondence tensor and, through the existing CM/LM calculus, a formula-valued logical tensor. The resulting framework provides a systematic bridge from continuous functions to evolving propositional operators while keeping continuous scores, Boolean truth, numeric correspondence operators, and symbolic logical matrices mathematically distinct.

Claim boundary. Threshold logic, sign abstraction, predicate abstraction, quantitative/fuzzy semantics, polynomial threshold functions, semialgebraic sign conditions, chamber geometry, and the basic theory of quasianalytic/non-quasianalytic Denjoy–Carleman classes all have substantial antecedents. The strongest result whose historical priority still requires dedicated verification is the Boolean-realizability dichotomy in Section 6: non-quasianalytic Denjoy–Carleman universality versus quasianalytic affine-parity rigidity under global whole-orthant exactness. The class-theoretic ingredients are standard; the claim under audit is their exact Boolean orthant classification and CM/LM use.

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1 Introduction

Continuous systems often support discrete descriptions. A temperature, voltage, activation score, control signal, or other real-valued quantity can vary continuously while propositions such as “above threshold,” “safe,” or “active” are binary. The passage from a real score to a Boolean proposition is classical: threshold devices and all-or-none neural models go back at least to McCulloch and Pitts [1], fuzzy and many-valued approaches retain intermediate values rather than immediately crisping them [2], and modern verification uses quantitative semantics whose signs recover Boolean satisfaction [5].

The present question is more specific. Suppose several continuous variables are thresholded into propositions. When do those propositions determine an exact Boolean *operator* for another continuous output, rather than merely supply individual binary labels? When the answer changes as continuous magnitudes or parameters vary, can that change itself be represented as an evolving family of logical operators?

The author’s earlier notes titled *Evolutionary Propositional Logic* explored this question through the logistic function, half-thresholding, and the identity that the sign of a product expresses Boolean equivalence. Those notes also proposed an “Evolutionary Logical Matrix” but mixed correct threshold identities with speculative interpretations [17]. The present paper keeps the valid kernel while replacing the tentative parts with a typed continuous-to-Boolean construction.

The endpoint is the companion CM/LM framework [16]. In that calculus a Boolean connective Θ , its numeric correspondence matrix $[\Theta]$, and its formula-valued logical matrix are distinct objects. The CM stores truth values in a declared operand frame, while the LM packages the polarity orbit of the corresponding formula and lifts to arbitrary arity. This paper asks the complementary question:

From which continuous systems can an exact Boolean operator Θ be extracted?

The answer is organized around the pipeline in Figure 1.



Figure 1: Typed continuous-to-Boolean-to-CM/LM pipeline. The link/score level and the Boolean operator level are intentionally kept distinct.

The main contributions are as follows.

1. An exact factorization criterion characterizes when a finite collection of Boolean predicates determines a genuine output connective.
2. Every continuous nonvanishing map $F : (\mathbb{R}^\times)^n \rightarrow \mathbb{R}^\times$ induces a unique Boolean function on the sign orthants, and continuous composition descends to Boolean composition.
3. Every Boolean function admits both a continuous piecewise-linear global orthant realization and an exact C^∞ realization.
4. More generally, every standard non-quasianalytic Denjoy–Carleman class admits exact realizations of all Boolean functions, while quasianalytic division classes are rigid under global nonvanishing: only affine-parity Boolean functions occur. Real-analytic and polynomial rigidity are corollaries, and the Gevrey scale exhibits the boundary explicitly at order 1.

5. For functions that are not globally sign-stable, continuous polarity tensors produce parameter-dependent correspondence tensors. Their switching sets partition magnitude or parameter space into operator chambers.
6. Each exact chamber operator lifts canonically into the existing formula-valued CM/LM calculus, yielding piecewise-constant trajectories through logical-operator space.

1.1 Prior-art boundary

The paper is deliberately conservative about novelty. Sign abstraction is a foundational example in abstract interpretation; Cousot and Cousot use the rule of signs to illustrate both exact and imprecise abstract operations [3]. Continuous/hybrid systems have long been mapped to discrete abstractions [4], and quantitative temporal-logic semantics retain real-valued robustness while preserving Boolean truth by sign [5]. Polynomial threshold functions study signs of real polynomials on the Boolean cube, with strong connections to hyperplane arrangements [8]. Threshold-function chambers themselves have a geometric literature going back to Zuev [6, 7] and have recently been developed explicitly in chamber/facet language by Anthony [9]. Semialgebraic sign conditions and connected-component bounds are standard tools of real algebraic geometry [10, 11].

Accordingly, the paper does not claim novelty for thresholding, sign abstraction, min/max logical semantics, sign-condition chambers, one-bit transitions between adjacent threshold functions, quasianalytic continuation, coordinate division, or non-quasianalytic bump-function machinery. The main historical question left open here is whether the resulting whole-orthant Boolean classification—quasianalytic affine-parity rigidity versus non-quasianalytic universality—has appeared previously in an equivalent realizability form.

2 Preliminaries and notation

Let

$$\mathbb{B} = \{0, 1\}, \quad \Sigma = \{-1, +1\}, \quad \mathbb{R}^\times = \mathbb{R} \setminus \{0\}.$$

Define the sign-to-Boolean map

$$q : \mathbb{R}^\times \rightarrow \mathbb{B}, \quad q(x) = \begin{cases} 1, & x > 0, \\ 0, & x < 0, \end{cases}$$

and the inverse sign encoding

$$\varepsilon : \mathbb{B} \rightarrow \Sigma, \quad \varepsilon(b) = 2b - 1.$$

Thus $\varepsilon(1) = +1$ and $\varepsilon(0) = -1$.

For $\alpha = (\alpha_1, \dots, \alpha_n) \in \mathbb{B}^n$, write

$$O_\alpha = \{x \in (\mathbb{R}^\times)^n : q(x_i) = \alpha_i \text{ for all } i\} = \{\varepsilon(\alpha) \odot r : r \in \mathbb{R}_{>0}^n\},$$

where $\odot$ is coordinatewise multiplication. The sets O_α are the 2^n open orthants.

2.1 Correspondence conventions

For a binary Boolean connective Θ , we use the true-first CM convention of the companion paper:

$$[\Theta] = \begin{bmatrix} 1\Theta 1 & 1\Theta 0 \\ 0\Theta 1 & 0\Theta 0 \end{bmatrix}.$$

In particular,

$$[\Leftrightarrow] = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad [\Updownarrow] = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}.$$

For arbitrary arity, a Boolean function $\Theta : \mathbb{B}^n \rightarrow \mathbb{B}$ has correspondence tensor

$$C_\Theta[\alpha] = \Theta(\alpha), \quad \alpha \in \mathbb{B}^n.$$

The companion CM/LM paper defines a faithful formula-valued LM lift of the same truth tensor [16]. The present paper will use that lift only after the continuous abstraction has produced a well-defined Boolean function.

Remark 2.1 (Zero as a switching boundary). The value zero is not assigned either Boolean sign. This is structural rather than cosmetic. A two-valued negation has no fixed point, whereas the continuous involution $u \mapsto 1 - u$ has the fixed point $u = 1/2$. Treating the threshold boundary separately prevents a tie from being silently identified with either truth value.

3 Continuous propositions and link functions

Definition 3.1 (Thresholded continuous proposition). Let $f : D \rightarrow \mathbb{R}$ be a real-valued score. On the set where $f(z) \neq 0$, its sign proposition is

$$P_f(z) = q(f(z)).$$

Proposition 3.2 (Monotone-link invariance). Let $L : \mathbb{R} \rightarrow I \subseteq \mathbb{R}$ be strictly increasing, let $c = L(0)$, and define

$$\delta_c(u) = \begin{cases} 1, & u > c, \\ 0, & u < c. \end{cases}$$

Then for every $x \neq 0$,

$$\delta_c(L(x)) = q(x).$$

Proof. Strict monotonicity gives

$$x > 0 \iff L(x) > L(0) = c, \quad x < 0 \iff L(x) < L(0) = c.$$

The thresholded link therefore records exactly the sign proposition. □

Corollary 3.3 (Logistic realization). For

$$S(x) = \frac{1}{1 + e^{-x}},$$

one has $S(0) = 1/2$ and hence

$$\delta_{1/2}(S(x)) = q(x), \quad x \neq 0.$$

Moreover, since $S(-x) = 1 - S(x)$,

$$\delta_{1/2}(S(-x)) = \neg \delta_{1/2}(S(x)).$$

Remark 3.4 (What the link contributes). The Booleanized output of Proposition 3.2 is independent of the particular strictly monotone link. The continuous quantity $L(x)$ may nevertheless retain useful margin or score information. In particular, a logistic value is not treated as a probability unless a probabilistic model supplies that interpretation.

4 Exact Boolean abstraction

Definition 4.1 (Proposition abstraction). Let

$$P = (P_1, \dots, P_n) : D \rightarrow \mathbb{B}^n$$

be a finite family of Boolean predicates on a state space D , and let $R : D \rightarrow \mathbb{B}$ be an output predicate. We say that R *factors exactly through* P if there exists a Boolean function $\Theta : \mathbb{B}^n \rightarrow \mathbb{B}$ satisfying

$$R = \Theta \circ P.$$

Theorem 4.2 (Exact Boolean factorization). *An exact Boolean factorization exists if and only if*

$$P(z) = P(z') \implies R(z) = R(z')$$

for all $z, z' \in D$. If $P(D) = \mathbb{B}^n$, then Θ is unique.

Proof. If $R = \Theta \circ P$ and $P(z) = P(z')$, then

$$R(z) = \Theta(P(z)) = \Theta(P(z')) = R(z').$$

Conversely, assume R is constant on every fiber of P . For each reachable $\alpha \in P(D)$ choose some z_α satisfying $P(z_\alpha) = \alpha$ and define

$$\Theta(\alpha) = R(z_\alpha).$$

The fiber condition makes this definition independent of the chosen representative. Assign arbitrary values to unreachable $\alpha \notin P(D)$. Then $R(z) = \Theta(P(z))$ for every $z \in D$. If $P(D) = \mathbb{B}^n$, no arbitrary assignments remain, so Θ is unique. $\square$

Definition 4.3 (Reachable truth domain). The set

$$\mathcal{R}_P = P(D) \subseteq \mathbb{B}^n$$

is the *reachable truth domain*. If $\mathcal{R}_P \neq \mathbb{B}^n$, an exact abstraction naturally determines a partial Boolean function on $\mathcal{R}_P$ even when its extension to all of $\mathbb{B}^n$ is nonunique.

Remark 4.4 (Exactness versus approximation). Failure of Theorem 4.2 does not imply that the Boolean predicates are useless. It says only that the current predicate set loses information relevant to the output. Section 10 records this loss explicitly with partial correspondence tensors and uses it to motivate refinement.

5 Orthant abstraction and compositionality

Theorem 5.1 (Orthant-induced Boolean operator). *Let*

$$F : (\mathbb{R}^\times)^n \rightarrow \mathbb{R}^\times$$

be continuous. Then there exists a unique Boolean function

$$\Theta_F : \mathbb{B}^n \rightarrow \mathbb{B}$$

such that

$$q(F(x_1, \dots, x_n)) = \Theta_F(q(x_1), \dots, q(x_n))$$

for every $x \in (\mathbb{R}^\times)^n$.

Proof. Each orthant O_α is connected. Since F is continuous, $F(O_\alpha)$ is connected. Since F never vanishes, this connected image lies in

$$\mathbb{R}^\times = (-\infty, 0) \cup (0, \infty),$$

whose two connected components are the negative and positive half-lines. Hence F has constant sign on O_α . Define $\Theta_F(\alpha)$ to be that sign encoded by q . Every orthant is nonempty, so the function is uniquely determined. $\square$

Proposition 5.2 (Elementary continuous realizations). *For nonzero inputs,*

$$\begin{aligned} q(-x) &= \neg q(x), \\ q(\min(x, y)) &= q(x) \wedge q(y), \\ q(\max(x, y)) &= q(x) \vee q(y), \\ q(xy) &= q(x) \Leftrightarrow q(y), \\ q(-xy) &= q(x) \Downarrow q(y), \\ q(\max(-x, y)) &= q(x) \Rightarrow q(y). \end{aligned}$$

Proof. Negation reverses sign. A minimum is positive precisely when both arguments are positive, while a maximum is negative precisely when both arguments are negative. The product xy is positive precisely when the two inputs have the same sign, giving equivalence; multiplying by -1 complements that result, giving XOR. Finally, $\max(-x, y)$ is negative precisely when $-x < 0$ and $y < 0$, equivalently $x > 0$ and $y < 0$, the unique false case of implication. $\square$

Example 5.3 (The historical multiplication identity). The original logistic notes observed that

$$\delta_{1/2}(S(xy)) = 1$$

precisely when x and y have the same sign. Proposition 5.2 shows that the logistic layer is optional: the underlying structural identity is

$$q(xy) = q(x) \Leftrightarrow q(y),$$

and the corresponding CM is

$$[\Leftrightarrow] = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

Theorem 5.4 (Composition). *Suppose*

$$F_i : (\mathbb{R}^\times)^n \rightarrow \mathbb{R}^\times, \quad i = 1, \dots, m,$$

induces Θ_i under Theorem 5.1, and suppose

$$G : (\mathbb{R}^\times)^m \rightarrow \mathbb{R}^\times$$

induces Ψ . Then

$$H(x) = G(F_1(x), \dots, F_m(x))$$

induces

$$\Theta_H = \Psi(\Theta_1, \dots, \Theta_m).$$

Proof. For every x ,

$$\begin{aligned} q(H(x)) &= q(G(F_1(x), \dots, F_m(x))) \\ &= \Psi(q(F_1(x)), \dots, q(F_m(x))) \\ &= \Psi(\Theta_1(q(x)), \dots, \Theta_m(q(x))). \end{aligned}$$

This is exactly the Boolean superposition of the induced functions. $\square$

Remark 5.5 (Bridge to the LM superposition theorem). The companion CM/LM calculus proves that its formula-valued LM lift preserves arbitrary Boolean superposition [16]. Theorem 5.4 therefore supplies a compositional front end: continuous composition descends to Boolean composition, which then lifts coherently to the CM/LM levels.

6 Global realizability of Boolean functions

Theorem 5.1 extracts Boolean operators from continuous maps. We now ask the converse question: given a Boolean operator, how regular can a continuous realization be while remaining exact on entire orthants?

6.1 Continuous piecewise-linear universality

Theorem 6.1 (Piecewise-linear universality). *Let $n \geq 1$. Every Boolean function*

$$\Theta : \mathbb{B}^n \rightarrow \mathbb{B}$$

has a continuous, positively homogeneous, piecewise-linear realization

$$F_\Theta : \mathbb{R}^n \rightarrow \mathbb{R}$$

such that

$$F_\Theta(x) \neq 0 \quad \text{whenever } x_i \neq 0 \text{ for all } i,$$

and

$$q(F_\Theta(x)) = \Theta(q(x_1), \dots, q(x_n)).$$

Proof. Every Boolean function can be expressed with $\neg$, $\wedge$, and $\vee$ (the two constant functions are handled separately below). Starting from the atomic realization $X_i \mapsto x_i$, recursively make the substitutions

$$\neg A \mapsto -F_A, \quad A \wedge B \mapsto \min(F_A, F_B), \quad A \vee B \mapsto \max(F_A, F_B).$$

Proposition 5.2 shows by structural induction that the sign realizes the intended Boolean formula. If $a, b \neq 0$, then $\min(a, b)$ and $\max(a, b)$ equal one of a, b and hence remain nonzero; negation also preserves nonzeroness. Thus no zero is introduced inside an open orthant. The operations $-$, $\min$, and $\max$ preserve continuity, piecewise linearity, and positive homogeneity under finite composition.

For the constant functions, one may take

$$F_\top(x) = \max(x_1, -x_1) = |x_1|, \quad F_\perp(x) = \min(x_1, -x_1) = -|x_1|,$$

which are respectively positive and negative whenever $x_1 \neq 0$. $\square$

Example 6.2 (Implication). A particularly compact realization is

$$F_{X \Rightarrow Y}(x, y) = \max(-x, y).$$

It is negative only when $x > 0$ and $y < 0$, so its induced CM is

$$[\Rightarrow] = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}.$$

6.2 Smooth universality

Define the one-sided flat functions

$$\phi_+(t) = \begin{cases} e^{-1/t^2}, & t > 0, \\ 0, & t \leq 0, \end{cases} \quad \phi_-(t) = \begin{cases} e^{-1/t^2}, & t < 0, \\ 0, & t \geq 0. \end{cases}$$

Both are C^∞ and all derivatives vanish at the origin.

Theorem 6.3 (C^∞ universality). *Every Boolean function $\Theta : \mathbb{B}^n \rightarrow \mathbb{B}$ admits an exact C^∞ orthant realization $F_\Theta : \mathbb{R}^n \rightarrow \mathbb{R}$.*

Proof. Encode the Boolean output as a sign function

$$h_\Theta(s) = 2\Theta\left(\frac{s_1+1}{2}, \dots, \frac{s_n+1}{2}\right) - 1, \quad s \in \Sigma^n.$$

For $s = (s_1, \dots, s_n)$ define

$$\Phi_s(x) = \prod_{i=1}^n \phi_{s_i}(x_i),$$

where $\phi_{+1} = \phi_+$ and $\phi_{-1} = \phi_-$. Set

$$F_\Theta(x) = \sum_{s \in \Sigma^n} h_\Theta(s) \Phi_s(x).$$

The sum is finite and each summand is C^∞ , so F_Θ is C^∞ . If $x \in (\mathbb{R}^\times)^n$ and $s = \text{sgn } x$, then every term $\Phi_{s'}(x)$ with $s' \neq s$ vanishes because at least one factor is evaluated on the wrong half-line. The unique surviving term is

$$F_\Theta(x) = h_\Theta(s) \prod_{i=1}^n e^{-1/x_i^2}.$$

The product is strictly positive, so $\text{sgn } F_\Theta(x) = h_\Theta(s)$, which is equivalent to the required Boolean identity. $\square$

Remark 6.4. The construction is infinitely differentiable but deliberately nonanalytic on the switching hyperplanes. This is not a technical accident: analyticity imposes the rigidity proved next.

6.3 Quasianalytic rigidity

The real-analytic argument uses less than analyticity itself. What matters is that the function class does not contain nontrivial flat germs and that vanishing on a coordinate hyperplane permits division by that coordinate. These properties are standard in important quasianalytic Denjoy–Carleman settings; see Bierstone and Milman [13] and Rainer [14, 15]. To keep the logical theorem independent of one particular weight-sequence convention, we isolate exactly the axioms used by the proof.

Definition 6.5 (Quasianalytic division class). A family $\mathcal{Q}(U) \subseteq C^\infty(U)$, indexed by open sets $U \subseteq \mathbb{R}^n$, will be called a *quasianalytic division class* for the purposes of this paper if it is a ring, is stable under ordinary restrictions, coordinate reflections, and restrictions to coordinate hyperplanes (with the induced lower-dimensional class), and has the following two properties.

(Q1) **Quasianalyticity.** On connected U , if the full Taylor jet of $f \in \mathcal{Q}(U)$ vanishes at one point, then $f \equiv 0$ on U .

(Q2) **Coordinate division.** If $f \in \mathcal{Q}(\mathbb{R}^n)$ vanishes on the full coordinate hyperplane $x_i = 0$, then $f = x_i g$ for some $g \in \mathcal{Q}(\mathbb{R}^n)$.

Real-analytic functions and the standard quasianalytic Denjoy–Carleman classes used in resolution theory are principal examples. The terminology here is deliberately descriptive rather than a proposal for a new function class.

Lemma 6.6 (Finite coordinate order). *Let $F \in \mathcal{Q}(\mathbb{R}^n)$ be nonzero, where $\mathcal{Q}$ is a quasianalytic division class. For every coordinate i there is a finite integer $m_i \geq 0$ and $G_i \in \mathcal{Q}(\mathbb{R}^n)$ such that*

$$F(x) = x_i^{m_i} G_i(x)$$

and the restriction $G_i|_{x_i=0}$ is not identically zero.

Proof. If $F|_{x_i=0}$ is not identically zero, take $m_i = 0$. Otherwise coordinate division gives $F = x_i F_1$. If F_1 also vanishes identically on the hyperplane, divide again. Suppose this process never stopped. Then F would be divisible by x_i^m for every m . Consequently every derivative of F at the origin vanishes: derivatives with x_i -order below m vanish because of the factor x_i^m , and m can be chosen larger than the x_i -order of any prescribed mixed derivative. Quasianalyticity would then force $F \equiv 0$, a contradiction. Hence the process stops after a finite m_i . $\square$

Lemma 6.7 (Uniform crossing parity). *Let $F \in \mathcal{Q}(\mathbb{R}^n)$ satisfy*

$$F(x) \neq 0 \quad \text{whenever } x \in (\mathbb{R}^\times)^n.$$

Fix a coordinate i . Then there exists $e_i \in \{0, 1\}$ such that, for every choice of signs of the other $n - 1$ coordinates, crossing $x_i = 0$ changes the sign of F by the same factor $(-1)^{e_i}$.

Proof. For $n = 1$ the assertion follows immediately from the finite order of vanishing at the origin, so assume $n \geq 2$. By Lemma 6.6, write

$$F = x_i^{m_i} G_i, \quad G_i|_{x_i=0} \neq 0.$$

Let V be any open orthant in the hyperplane $x_i = 0$, determined by the signs of the remaining coordinates. The restriction of G_i to that hyperplane belongs to the corresponding quasianalytic class. If it vanished on all of V , it would vanish on a nonempty open subset and hence have zero

Taylor jet at every point of that subset; quasianalyticity on the connected hyperplane would force the restriction to vanish identically, contrary to construction. Thus choose $a \in V$ with $G_i(0, a) \neq 0$.

For sufficiently small $t > 0$, continuity gives

$$\operatorname{sgn} F(t, a) = \operatorname{sgn} G_i(0, a), \quad \operatorname{sgn} F(-t, a) = (-1)^{m_i} \operatorname{sgn} G_i(0, a).$$

Because F is nonzero throughout each open orthant, its sign is constant on each of the two full orthants adjacent to V . Their sign ratio is therefore $(-1)^{m_i}$. Since the same m_i works for every V , the crossing factor is independent of all other input signs. Put $e_i = m_i \bmod 2$. $\square$

Theorem 6.8 (Global quasianalytic orthant rigidity). *Let $F \in \mathcal{Q}(\mathbb{R}^n)$ belong to a quasianalytic division class and suppose*

$$F(x) \neq 0 \quad \text{for every } x \in (\mathbb{R}^\times)^n.$$

Then there exist $c \in \{-1, +1\}$ and $e_1, \dots, e_n \in \{0, 1\}$ such that

$$\operatorname{sgn} F(x) = c \prod_{i=1}^n \operatorname{sgn}(x_i)^{e_i}$$

for every $x \in (\mathbb{R}^\times)^n$. Consequently, the induced Boolean function is affine parity. Conversely, every affine-parity sign law is realized by a polynomial monomial.

Proof. By Theorem 5.1, F has constant sign on every orthant. Let c be its sign on the positive orthant. Lemma 6.7 says that flipping coordinate i multiplies the output sign by one fixed factor $(-1)^{e_i}$, independent of all other signs. Starting from the positive orthant and flipping precisely those coordinates that are negative at x yields the displayed formula. Under the Boolean encoding this is an XOR of a subset of the input bits, possibly followed by output complement, hence an affine Boolean function over $\mathbb{F}_2$.

Conversely,

$$P(x) = c \prod_{i:e_i=1} x_i$$

is polynomial, nonzero on $(\mathbb{R}^\times)^n$, and has exactly the required sign law. $\square$

Corollary 6.9 (Global real-analytic orthant rigidity). *A real-analytic $F : \mathbb{R}^n \rightarrow \mathbb{R}$ that is nonzero on $(\mathbb{R}^\times)^n$ induces an affine-parity Boolean function, and every affine-parity Boolean function has a polynomial realization.*

Corollary 6.10 (Polynomial rigidity). *A polynomial that is nonzero on every open orthant can induce only an affine-parity Boolean function. Conversely every affine-parity Boolean function has such a polynomial realization.*

6.4 Non-quasianalytic Denjoy–Carleman universality

The converse phenomenon is supplied by flat functions. For a standard Denjoy–Carleman class C^M associated with a DC-weight sequence, non-quasianalyticity is equivalent to the existence of nontrivial flat functions and yields C^M partitions of unity. Rainer records an especially useful one-sided form: in a non-quasianalytic class one may choose $g \in C^M(\mathbb{R})$ with

$$g(t) = 0 \quad (t \leq 0),$$

while g is not identically zero in any interval $(0, \varepsilon)$ [14]. We turn this standard ingredient into an orthant selector.

Lemma 6.11 (Positive one-sided selector). *Let C^M be a non-quasianalytic Denjoy–Carleman class of the standard differential-ring type. Then there is $\psi_+ \in C^M(\mathbb{R})$ such that*

$$\psi_+(t) = 0 \quad (t \leq 0), \quad \psi_+(t) > 0 \quad (t > 0).$$

The reflected function $\psi_-(t) = \psi_+(-t)$ has the opposite support property.

Proof. Take the one-sided flat function g described above and define

$$\psi_+(t) = \int_0^t g(s)^2 ds.$$

For $t \leq 0$ the integrand vanishes on the integration interval, so $\psi_+(t) = 0$. For every $t > 0$, the hypothesis that g is nonzero in every right neighborhood of 0 implies that g^2 is positive somewhere in $(0, t)$, hence $\psi_+(t) > 0$. Standard Denjoy–Carleman derivative estimates are preserved under finite products and one antiderivative: for derivative order $k \geq 1$, the k th derivative of ψ_+ is the $(k-1)$ st derivative of g^2 , while the zeroth-order term is locally bounded. Hence ψ_+ remains in C^M ; equivalently this is consistent with the standard partition-of-unity construction in the non-quasianalytic class [14, 15]. $\square$

Theorem 6.12 (Non-quasianalytic Denjoy–Carleman universality). *Let C^M be a standard non-quasianalytic Denjoy–Carleman class. Every Boolean function $\Theta : \mathbb{B}^n \rightarrow \mathbb{B}$ has an exact global orthant realization $F_\Theta \in C^M(\mathbb{R}^n)$ satisfying*

$$F_\Theta(x) \neq 0 \quad (x \in (\mathbb{R}^\times)^n)$$

and

$$q(F_\Theta(x)) = \Theta(q(x_1), \dots, q(x_n)).$$

Proof. Let $h_\Theta : \Sigma^n \rightarrow \{-1, +1\}$ be the sign encoding of Θ . With the selectors from Lemma 6.11, define for $s \in \Sigma^n$

$$\Psi_s(x) = \prod_{i=1}^n \psi_{s_i}(x_i), \quad \psi_{+1} = \psi_+, \quad \psi_{-1} = \psi_-.$$

Then set

$$F_\Theta(x) = \sum_{s \in \Sigma^n} h_\Theta(s) \Psi_s(x).$$

The class is a ring and is stable under coordinate reflections, so $F_\Theta \in C^M$. If $x \in (\mathbb{R}^\times)^n$ has sign vector s , every term with index $s' \neq s$ contains at least one selector evaluated on its zero half-line and vanishes. The unique surviving term is

$$F_\Theta(x) = h_\Theta(s) \prod_i \psi_{s_i}(x_i),$$

whose product factor is strictly positive. Thus F_Θ is nonzero and has the required Boolean sign on every open orthant. $\square$

Corollary 6.13 (Denjoy–Carleman Boolean dichotomy). *For standard Denjoy–Carleman classes with the usual ring, composition/reflection, and coordinate-division properties, whole-orthant Boolean realizability has the following dichotomy:*

- (i) *if the class is quasianalytic, the globally exact nonvanishing sign operators are exactly the affine-parity functions;*

(ii) if the class is non-quasianalytic, every Boolean function is globally exactly realizable.

Corollary 6.14 (Gevrey phase transition). *In the standard Gevrey scale G^s , the whole-orthant Boolean expressivity changes sharply at $s = 1$: $G^1 = C^\omega$ admits only affine-parity operators under global nonvanishing, whereas every G^s with $s > 1$ admits all Boolean functions.*

Proof. The class G^1 is real analytic. For $s > 1$, G^s is non-quasianalytic and contains one-sided flat functions of the familiar form

$$\phi_s(t) = \begin{cases} \exp(-t^{-1/(s-1)}), & t > 0, \\ 0, & t \leq 0, \end{cases}$$

from which the preceding selector construction may be carried out. $\square$

Corollary 6.15 (Counting quasianalytic sign operators). *Among the 2^{2^n} Boolean functions of n variables, exactly 2^{n+1} occur as globally exact sign operators in a quasianalytic division class under the nonvanishing hypothesis.*

Corollary 6.16 (Binary case). *For two variables the globally quasianalytic realizable connectives are exactly*

$$\perp, \top, X, \neg X, Y, \neg Y, X \Leftrightarrow Y, X \Updownarrow Y.$$

The remaining eight binary Boolean operators have exact realizations in every standard non-quasianalytic Denjoy–Carleman class by Theorem 6.12, but no realization in a quasianalytic division class satisfying the stated global nonvanishing condition.

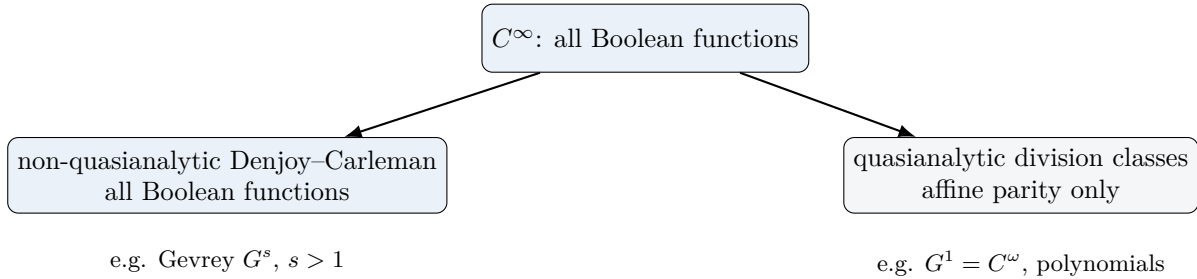


Figure 2: Regularity boundary for exact whole-orthant Boolean realization. In standard Denjoy–Carleman scales, the decisive distinction is non-quasianalytic versus quasianalytic, not merely smooth versus real analytic.

6.5 Priority status of the regularity dichotomy

The ingredients of Corollary 6.13 are classical: quasianalytic classes are characterized by exclusion of nontrivial flat germs, coordinate-division properties are standard in the classes used in quasianalytic geometry, and non-quasianalytic Denjoy–Carleman classes admit flat functions and partitions of unity [13, 14, 15]. The polynomial special case of the crossing-parity obstruction is also elementary and closely related to older sign-separation arguments. The contribution under investigation is therefore not any of these ingredients in isolation, but the resulting exact Boolean classification on whole orthants and its use as the regularity boundary of the CM/LM abstraction pipeline.

A targeted literature search conducted for this revision found extensive work on quasianalyticity, Denjoy–Carleman division, polynomial threshold functions, and sign-region geometry, but did not locate an exact source packaging the Boolean orthant dichotomy of Corollary 6.13. This is evidence for further priority investigation, not proof of novelty. We therefore retain conservative language: Theorem 6.8, Theorem 6.12, and their dichotomy are treated as candidate results pending specialist review.

7 Continuous polarity tensors and CM extraction

Global orthant exactness is only one possibility. A continuous function may induce different Boolean operators at different magnitudes or parameter values. To represent this, we separate sign pattern from magnitude.

Definition 7.1 (Continuous polarity score tensor). Let $F : \mathbb{R}^n \rightarrow \mathbb{R}$, let $r = (r_1, \dots, r_n) \in \mathbb{R}_{>0}^n$, and let L be a strictly monotone link. For each $\alpha \in \mathbb{B}^n$ define

$$x_\alpha(r) = \varepsilon(\alpha) \odot r.$$

The *continuous polarity score tensor* is

$$\mathcal{S}_{F,L}(r)[\alpha] = L(F(x_\alpha(r))).$$

Definition 7.2 (Induced correspondence tensor). Whenever $F(x_\alpha(r)) \neq 0$ for every α , define

$$C_F(r)[\alpha] = q(F(x_\alpha(r))).$$

For $n = 2$ this is the true-first matrix

$$C_F(a, b) = \begin{bmatrix} q(F(a, b)) & q(F(a, -b)) \\ q(F(-a, b)) & q(F(-a, -b)) \end{bmatrix}.$$

Example 7.3 (Multiplication gives a fixed CM). For $F(x, y) = xy$ and the logistic link,

$$\mathcal{S}_{F,S}(a, b) = \begin{bmatrix} S(ab) & S(-ab) \\ S(-ab) & S(ab) \end{bmatrix}.$$

Thresholding entrywise gives

$$C_F(a, b) = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = [\Leftrightarrow]$$

for every $a, b > 0$.

Proposition 7.4 (Fixed-CM criterion). *Suppose $F(x) \neq 0$ for every $x \in (\mathbb{R}^\times)^n$. The following are equivalent:*

- (i) F induces one global Boolean operator under Theorem 5.1;
- (ii) $C_F(r)$ is independent of $r \in \mathbb{R}_{>0}^n$;
- (iii) for every α , the sign of $F(\varepsilon(\alpha) \odot r)$ is independent of r .

Proof. If a global operator Θ_F exists then every tensor entry is $\Theta_F(\alpha)$, proving (i) $\Rightarrow$ (ii). Equality of tensors implies equality entrywise, giving (ii) $\Rightarrow$ (iii). Under (iii), define $\Theta_F(\alpha)$ to be the common sign of $F(\varepsilon(\alpha) \odot r)$; this is well-defined and proves (iii) $\Rightarrow$ (i). $\square$

8 Evolving operators and operator chambers

For $\alpha \in \mathbb{B}^n$, define the polarity score

$$g_\alpha(r) = F(\varepsilon(\alpha) \odot r), \quad r \in \mathbb{R}_{>0}^n.$$

Definition 8.1 (Discriminant and operator chamber). Define

$$\Delta_F = \bigcup_{\alpha \in \mathbb{B}^n} \{r \in \mathbb{R}_{>0}^n : g_\alpha(r) = 0\}.$$

A connected component of

$$\mathbb{R}_{>0}^n \setminus \Delta_F$$

is called an *operator chamber*.

Theorem 8.2 (Operator-chamber theorem). *The map*

$$r \mapsto C_F(r)$$

is constant on every operator chamber.

Proof. Let U be a connected component of $\mathbb{R}_{>0}^n \setminus \Delta_F$. For each α , the continuous function g_α never vanishes on U . Hence $g_\alpha(U)$ is connected and lies in $\mathbb{R}^\times$, so it is contained wholly in either $(0, \infty)$ or $(-\infty, 0)$. Thus $q(g_\alpha(r))$ is constant on U for every α . Every entry of $C_F(r)$ is therefore constant on U . $\square$

Corollary 8.3 (Finite polynomial operator atlas). *If F is polynomial, the operator-chamber decomposition has finitely many semialgebraically connected chambers.*

Proof. Each g_α is polynomial. The chambers are connected components of realizations of strict sign conditions of the finite polynomial family $\{g_\alpha\}$. Such realizations are semialgebraic, and semialgebraic sets have finitely many connected components [10, 11]. $\square$

Proposition 8.4 (Generic single-entry transition). *Let $\gamma : (-\epsilon, \epsilon) \rightarrow \mathbb{R}_{>0}^n$ be continuous. Suppose exactly one polarity score $g_{\alpha_0}(\gamma(t))$ crosses zero at $t = 0$, while all other polarity scores remain nonzero near 0. Then the correspondence tensors immediately before and after the crossing differ in exactly the entry α_0 .*

Proof. For $\alpha \neq \alpha_0$, continuity and local nonvanishing force $q(g_\alpha(\gamma(t)))$ to remain constant across $t = 0$. By assumption, $g_{\alpha_0} \circ \gamma$ changes sign, so precisely that one truth-table entry flips. $\square$

Remark 8.5 (Prior art for chamber adjacency). For linear threshold functions, parameter space is already known to decompose into hyperplane-arrangement chambers corresponding to Boolean threshold functions; chamber facets and one-input changes are closely related in work of Zuev and in Anthony's recent geometric formulation [6, 7, 9]. Proposition 8.4 is therefore not claimed as a new threshold-logic phenomenon. Its role here is to express the same local geometry directly in CM coordinates and to extend the representation to general continuous polarity families.

Proposition 8.6 (One-parameter polynomial switching bound). *Suppose a one-parameter family has polarity scores $g_\alpha(t)$, $\alpha \in \mathbb{B}^n$, where every g_α is a nonzero polynomial of degree at most d . Then at most 2^nd switching parameter values occur and the induced Boolean operator is constant on at most $2^nd + 1$ open parameter intervals.*

Proof. Each nonzero g_α has at most d real roots. The union of the zero sets of all 2^n polarity polynomials therefore has at most 2^nd points. Removing N points from the real line leaves at most $N + 1$ open intervals, and Theorem 8.2 makes the CM constant on each interval. $\square$

9 Worked operator evolutions

9.1 Addition: magnitude determines the operator

Let

$$F(x, y) = x + y.$$

For positive magnitudes a, b ,

$$C_F(a, b) = \begin{bmatrix} 1 & q(a - b) \\ q(-a + b) & 0 \end{bmatrix}.$$

Hence

$$a > b \implies C_F(a, b) = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} = [X],$$

while

$$a < b \implies C_F(a, b) = \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix} = [Y].$$

At $a = b$ the two mixed-polarity scores reach zero. Thus addition does not have a single global sign-CM; it realizes the evolving operator

$$[X] \longleftrightarrow [Y].$$

This is the same information loss highlighted by the classical rule-of-signs abstraction: opposite-sign addition cannot be decided from signs alone [3].

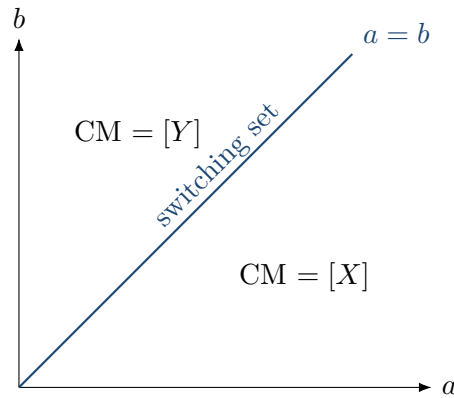


Figure 3: Operator chambers for addition in positive magnitude space. The same continuous function induces different Boolean projections depending on whether $|x|$ or $|y|$ dominates.

9.2 An affine threshold gate that evolves through five connectives

Consider

$$F_c(x, y) = 2x + y + c$$

at unit magnitudes. The true-first polarity scores are

$$c + 3, \quad c + 1, \quad c - 1, \quad c - 3.$$

Therefore

$$C_{F_c} = \begin{cases} [\perp], & c < -3, \\ [\wedge], & -3 < c < -1, \\ [X], & -1 < c < 1, \\ [\vee], & 1 < c < 3, \\ [\top], & c > 3. \end{cases}$$

Thus a continuously varying bias produces the discrete operator path

$$\boxed{\perp \longrightarrow \wedge \longrightarrow X \longrightarrow \vee \longrightarrow \top}.$$

At each generic boundary exactly one truth-table entry changes, in agreement with Proposition 8.4.

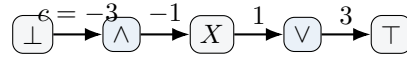


Figure 4: One-dimensional operator evolution for $F_c(x, y) = 2x + y + c$ at unit magnitudes.

9.3 Fixed versus evolving functions

Multiplication and addition provide a useful contrast:

$$xy \rightsquigarrow [\Leftrightarrow] \text{ for all positive magnitudes,}$$

whereas

$$x + y \rightsquigarrow [X] \text{ or } [Y]$$

depending on the magnitude chamber. The distinction is not whether a continuous function can be thresholded; both can. The distinction is whether the resulting truth tensor is invariant over the continuous cells represented by the chosen Boolean abstraction.

10 Partial correspondence tensors and refinement

When an abstraction is not exact, forcing a 0 or 1 into every CM cell hides the failure. A useful intermediate representation records uncertainty explicitly.

Definition 10.1 (Partial correspondence value). For an abstract input cell, use the four symbols

$$\{0, 1, \star, \perp\},$$

where 0 means definitely false, 1 definitely true, $\star$ means both Boolean outputs occur within the cell, and $\perp$ means that the cell is unreachable.

Definition 10.2 (Partial correspondence tensor). Given predicates $P_1, \dots, P_n$ and an output predicate R , define

$$\hat{C}[\alpha] = \begin{cases} \perp, & P^{-1}(\alpha) = \emptyset, \\ 0, & R = 0 \text{ throughout } P^{-1}(\alpha), \\ 1, & R = 1 \text{ throughout } P^{-1}(\alpha), \\ \star, & \text{both outputs occur in } P^{-1}(\alpha). \end{cases}$$

Then $\hat{C}$ is exact precisely when no reachable entry is $\star$.

Example 10.3 (Refining addition). For $R = q(x + y)$, the two predicates $X = q(x)$ and $Y = q(y)$ are insufficient in the opposite-sign cells. Adding

$$M = (|x| > |y|)$$

resolves the ambiguity away from $|x| = |y|$. Thus an ambiguous CM cell can identify the missing discriminatory information required for an exact Boolean abstraction.

Algorithm 10.4 (Exact CM extraction). Given a continuous model and predicates $P_1, \dots, P_n$:

1. construct each Boolean predicate cell $P^{-1}(\alpha)$;
2. test whether the output can be positive there;
3. test whether it can be negative there;
4. record 1, 0, $\star$, or $\perp$;
5. if no reachable $\star$ remains, emit the exact correspondence tensor;
6. otherwise refine the predicate set or retain the partial tensor explicitly.

Predicate refinement is established in verification and abstraction theory [4, 3]. The CM-facing contribution proposed here is to make the target object an explicit truth tensor/CM that can subsequently enter the LM calculus.

11 Lifting operator chambers into the CM/LM calculus

Theorem 11.1 (Chamber-to-CM lift). *Let U be an operator chamber of F . Then there is a unique Boolean function Θ_U satisfying*

$$C_F(r) = C_{\Theta_U} \quad \text{for every } r \in U.$$

Proof. Theorem 8.2 makes $C_F(r)$ constant on U . A truth tensor uniquely determines a Boolean function on its declared input frame, so the common tensor is C_{Θ_U} for a unique Θ_U . $\square$

Let L_X denote the formula-valued LM lift of the companion CM/LM calculus [16]. The chamber therefore determines the typed chain

$$\boxed{U \longmapsto \Theta_U \longmapsto C_{\Theta_U} \longmapsto L_X(\Theta_U).}$$

The companion theory proves that this lift is faithful and preserves pointwise Boolean superposition. Hence the present continuous abstraction can feed directly into a symbolic logical-operator layer without identifying the continuous score itself with an LM.

Corollary 11.2 (Evolving LM trajectory). *Let $r : [0, T] \rightarrow \mathbb{R}_{>0}^n$ be a continuous parameter trajectory meeting the discriminant only at isolated times. Then*

$$t \longmapsto L_X(\Theta_{r(t)})$$

is piecewise constant.

Remark 11.3 (Typed separation). Five objects should remain distinct:

$$F, \quad \mathcal{S}_F, \quad \Theta_F, \quad C_F, \quad L_X(\Theta_F).$$

They are respectively a continuous function, a continuous score tensor, a Boolean function, a numeric correspondence tensor, and a formula-valued logical tensor. Much of the conceptual clarity of the construction comes from not collapsing these levels into one another.

12 Computational interpretation

The formalism suggests a direct compilation workflow from continuous models to guarded logical operators.

Algorithm 12.1 (Operator-chamber construction). For a parameterized continuous model F_r :

1. construct the polarity scores $g_\alpha(r) = F_r(\varepsilon(\alpha) \odot m)$ for the chosen magnitude/input convention;
2. compute, solve, or approximate their zero sets;
3. partition parameter space by the resulting sign conditions;
4. evaluate one sample point per connected chamber;
5. assign its truth tensor/CM;
6. optionally merge adjacent chambers carrying the same Boolean operator;
7. lift each surviving operator into the formula-valued LM representation.

For polynomial families, cylindrical algebraic decomposition and other real-algebraic sign-decomposition methods give exact routes in principle, although the complexity can be high. For neural or black-box models, interval methods, SMT solving, abstract interpretation, or verified optimization could instead be used to certify signs within cells. The framework itself is agnostic to the certification method: the required invariant is simply that each emitted Boolean cell has a proved output sign.

A useful final representation is a guarded operator atlas

$$\text{continuous chamber condition} \implies \text{CM/LM operator}.$$

This makes explicit both the continuous regime in which an operator applies and the discrete operator that regime implements.

13 Relationship to prior work and claim boundary

13.1 Threshold logic and neural activation

McCulloch and Pitts explicitly connected all-or-none neural activity with propositional logic [1]. Modern threshold and neural models routinely compute a real score followed by a discrete decision. Proposition 3.2 therefore organizes a standard observation rather than claiming a new activation principle.

13.2 Fuzzy and quantitative semantics

Zadeh’s fuzzy sets introduced continuum-valued membership [2]. In formal verification, robustness semantics assign signed real values whose sign recovers Boolean satisfaction; min/max and real negation are standard quantitative counterparts of conjunction, disjunction, and negation in important fragments [5]. The elementary realizations in Proposition 5.2 should therefore be read as compatibility with established continuous semantics.

13.3 Abstract interpretation and sign abstraction

Cousot and Cousot’s foundational abstract-interpretation paper uses the rule of signs to show how concrete arithmetic can be interpreted over a finite abstract domain [3]. Multiplication is often exact at the sign level while mixed-sign addition is imprecise. The fixed-CM/evolving-CM distinction in this paper refines that observation into an explicit truth-tensor representation; the underlying exactness issue is classical.

13.4 Discrete abstractions of continuous and hybrid systems

Discrete quotients of hybrid and continuous systems are well established [4]. Theorem 4.2 is a minimal set-theoretic exactness criterion rather than a new theory of abstraction. Its role is to make the CM/LM interface precise.

13.5 Quasianalytic and Denjoy–Carleman regularity

Quasianalytic Denjoy–Carleman classes generalize real-analytic functions while retaining an injective Taylor map and hence an analytic-continuation-type principle. Bierstone and Milman develop quasianalytic classes with the differential-calculus and division properties needed in resolution theory [13]. On the other side of the boundary, Rainer records that a non-quasianalytic Denjoy–Carleman class admits partitions of unity and, in particular, one-sided flat functions from which bump functions can be constructed [14]; a modern survey develops the broader ultradifferentiable extension framework and the role of (non-)quasianalyticity [15]. These are direct antecedents for the function-class ingredients of Section 6. The candidate contribution is the Boolean whole-orthant classification obtained by combining those ingredients with the exact sign-realization problem.

13.6 Polynomial threshold functions

Polynomial threshold functions constrain the sign of a real polynomial on the discrete Boolean cube. Baldi and Vershynin connect their enumeration to hyperplane arrangements and random tensors [8].

The whole-orthant condition of Theorem 6.9 is substantially stronger: it requires the sign to be fixed at every point of an open orthant, not only at finitely many vertices.

13.7 Chamber geometry

Zuev’s work established geometric methods in threshold logic [6, 7]. Anthony’s recent preprint makes the chamber interpretation explicit: threshold functions correspond to chambers in weight/threshold parameter space, and chamber facets are tied to essential inputs and one-inclusion adjacency [9]. Consequently, the chamber idea and generic one-entry transition are not claimed as new. The present use is to label continuous magnitude/parameter regions directly by CM/LM operators, including families not restricted to ordinary threshold gates.

13.8 Real algebraic sign regions

For polynomial families, the discriminant and chamber decomposition are instances of finite sign-condition geometry. Bounds on connected components of realizable sign conditions are standard in real algebraic geometry [10, 11]. Corollary 8.3 is included because it supplies the exact finiteness statement needed for the operator atlas, not as a new semialgebraic theorem.

13.9 Candidate contribution table

14 Applications and use cases

The framework is intended to be useful even where the individual ingredients are classical.

Threshold and neural units. For a score $z = w^\top x + b$, varying weights, bias, or input magnitudes can change the Boolean operator induced on a declared set of input propositions. The operator atlas reports those changes explicitly rather than only classifying individual samples.

Analog and mixed-signal systems. Continuous voltages may implement digital relationships only within specified operating regions. A chamber-labelled CM makes the corresponding logic and its regime of validity explicit.

Hybrid control. Continuous trajectories can induce discrete transitions not merely among truth states but among logical relationships between thresholded observables. Guarded CM/LM operators provide one compact representation of those relationships.

Scientific classification rules. A continuous model may admit a finite exact abstraction over some variables while requiring additional predicates in others. Partial correspondence tensors expose exactly where the coarse abstraction fails.

Explainability. Given a continuous model, a natural question is not only “what label does this point receive?” but also “what Boolean relationship does this entire regime implement?” The CM/LM endpoint answers the latter with a reusable logical object.

Result or construction	Status here	Claim boundary
Monotone link + threshold	Established	Standard threshold/sign fact
Exact factorization through predicates	Established principle	Quotient/fiber criterion; abstraction antecedents
Sign multiplication $\rightarrow$ equivalence	Established	Rule-of-signs/sign algebra
PL realization of all Boolean functions	Synthesis theorem	Built from standard $-$, $\min$, $\max$ semantics
C^∞ realization of all Boolean functions	Explicit construction	Classical flat-function machinery; priority not asserted
Quasianalytic affine-parity rigidity	Candidate theorem/formulation	Uses standard quasianalyticity and coordinate division; exact Boolean classification not located
Non-quasianalytic Denjoy–Carleman universality	Candidate theorem/application	Flat functions and partitions of unity are standard; orthant realization is the derived claim
Denjoy–Carleman / Gevrey dichotomy	Strong candidate synthesis	Sharp Boolean expressivity boundary pending specialist priority review
Real-analytic and polynomial rigidity	Corollaries	Polynomial crossing-parity mechanism has strong prior-art pressure
Operator chambers for polynomial families	Established geometry	Sign conditions/real algebraic geometry
Generic one-entry CM transition	Established in threshold setting	Zuev/Anthony chamber adjacency
Continuous polarity tensor $\rightarrow$ CM/LM atlas	Architectural contribution	CM/LM-specific integration proposed here
Partial CM + refinement target	Application direction	Predicate refinement has strong antecedents

Table 1: Conservative claim-status summary.

15 Open problems

1. **Priority and generality of the quasianalytic dichotomy.** Determine whether Theorem 6.8, Theorem 6.12, or Corollary 6.13 appears in equivalent form in quasianalytic geometry, Denjoy–Carleman theory, sign-pattern theory, or threshold-function literature.
2. **Minimal axioms for rigidity.** Coordinate division and quasianalytic continuation suffice for the proof given here. Which assumptions can be weakened while preserving affine-parity rigidity? Do analogous results hold in Braun–Meise–Taylor or polynomially bounded o-minimal quasianalytic classes?
3. **Regularity phase diagrams.** The Gevrey scale exhibits a sharp change at $s = 1$. Determine corresponding expressivity boundaries for other ultradifferentiable scales and for local versus global orthant domains.
4. **Restricted domains.** Analytic rigidity uses whole open orthants. Which non-affine Boolean functions admit polynomial or analytic realizations on bounded magnitude boxes $a_i \leq |x_i| \leq b_i$?
5. **Minimal realization complexity.** For a given Boolean function, minimize the number of

min/max operations, smooth bump components, critical points, or switching surfaces required by an exact continuous realization.

6. **Minimal refinement.** Given a partial tensor containing $\star$ entries, find the smallest additional predicate family that produces an exact CM.
7. **Chamber complexity.** For a declared continuous function class, bound the number of distinct Boolean operators that can occur as parameters or magnitudes vary.
8. **Realizable transition graphs.** Characterize which paths or subgraphs of the Boolean-function hypercube can be generated by one-parameter continuous families.
9. **Reachability-constrained CMs.** Develop the natural CM/LM representation when only a strict subset of Boolean input assignments is physically reachable.
10. **Compiler integration.** Combine sign certification, chamber construction, CM extraction, and formula-valued LM lifting into a verified continuous-to-logic compiler.

16 Discussion

Three different kinds of change should be distinguished. First, a continuous score can vary while its truth value remains fixed. Second, the truth value of a proposition can switch when a threshold surface is crossed. Third, an entire Boolean *operator* can change when the relative magnitudes or parameters cause different polarity cases to cross different switching surfaces. The third phenomenon is the precise sense in which the present framework uses the term *evolving operator*.

The realizability hierarchy gives the paper a second theme. Smoothness alone is not restrictive enough to limit Boolean expressivity: every finite Boolean function has an exact C^∞ orthant realization. The deeper boundary is quasianalyticity. Non-quasianalytic Denjoy–Carleman classes contain flat one-sided selectors and therefore support arbitrary orthant sign prescriptions. Quasianalytic division classes exclude that flat freedom; coordinate division and unique continuation force a coordinate flip to have the same parity effect in every other orthant, collapsing globally exact truth functions to affine parity. Real analyticity is an important special case rather than the whole phenomenon.

This distinction also clarifies why the logistic function is useful but not foundational. A logistic or other monotone link can provide a smooth score representation of a sign before thresholding, but the Boolean operator is determined by the signs of the underlying polarity scores. The continuous score tensor and the thresholded CM are therefore complementary objects rather than competing representations.

Finally, the CM/LM connection gives the abstraction a typed endpoint. Once a chamber has been certified, its Boolean truth tensor is not merely a list of bits: it participates in the existing CM/LM operator calculus, including valuation, frame transport, formula-valued lifting, and Boolean superposition [16]. This makes the continuous construction potentially useful as an interface between real-valued models and symbolic operator computation.

17 Conclusion

We developed a framework for extracting exact Boolean operators from continuous functions and transporting those operators into correspondence-matrix and logical-matrix representations. The

exact-factorization criterion states precisely when a finite proposition abstraction determines an output operator. On open real orthants, every continuous nonvanishing function automatically induces a Boolean function, and continuous composition descends to Boolean composition.

Conversely, every Boolean function has an exact continuous piecewise-linear realization and an exact C^∞ realization. More sharply, standard non-quasianalytic Denjoy–Carleman classes realize every Boolean function, whereas quasianalytic division classes are rigid under global nonvanishing: their orthant sign laws are necessarily affine parity. The Gevrey scale makes this boundary explicit at order $s = 1$.

When a continuous function is not globally sign-stable, its polarity scores define switching surfaces that divide magnitude or parameter space into operator chambers. Each chamber carries a definite Boolean correspondence tensor and therefore a canonical formula-valued LM lift. Continuous trajectories can consequently induce piecewise-constant trajectories through spaces of Boolean operators, providing a mathematically precise interpretation of evolving propositions and evolving logical operators.

A All sixteen binary connectives as continuous realizations

Table 2 gives compact exact realizations for every binary Boolean function. The first eight marked “analytic” are exactly the affine-parity family from Corollary 6.16; the other eight are represented here with min/max constructions and cannot be realized in any quasianalytic division class under the global exactness hypothesis of Theorem 6.8.

Table 2: Exact continuous realizations of the sixteen binary Boolean operators in true-first CM order.

Operator	CM	Continuous realization $F(x, y)$	Global analytic?
$\perp$	$\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$	$-x^2$	yes
$\top$	$\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$	x^2	yes
X	$\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$	x	yes
$\neg X$	$\begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix}$	$-x$	yes
Y	$\begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}$	y	yes
$\neg Y$	$\begin{bmatrix} 0 & 1 \\ 0 & 1 \end{bmatrix}$	$-y$	yes
$X \Leftrightarrow Y$	$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$	xy	yes

Operator	CM	Continuous realization $F(x, y)$	Global analytic?
$X \Updownarrow Y$	$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$	$-xy$	yes
$X \wedge Y$	$\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$	$\min(x, y)$	no
$\neg(X \wedge Y)$	$\begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}$	$\max(-x, -y)$	no
$X \vee Y$	$\begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$	$\max(x, y)$	no
$\neg(X \vee Y)$	$\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$	$\min(-x, -y)$	no
$X \Rightarrow Y$	$\begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$	$\max(-x, y)$	no
$Y \Rightarrow X$	$\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$	$\max(x, -y)$	no
$X \wedge \neg Y$	$\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$	$\min(x, -y)$	no
$\neg X \wedge Y$	$\begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$	$\min(-x, y)$	no

Remark A.1. The “no” entries mean that no function in a quasianalytic division class satisfying the *global whole-orthant nonvanishing condition* of Theorem 6.8 can realize that connective. In particular there is no such real-analytic realization. This does not preclude analytic threshold representations on a discrete Boolean cube or on restricted bounded magnitude domains.

B Additional chamber examples

B.1 A shifted product

Consider

$$F_\lambda(x, y) = xy + \lambda$$

at unit magnitudes. The four polarity scores are

$$1 + \lambda, \quad -1 + \lambda, \quad -1 + \lambda, \quad 1 + \lambda.$$

Therefore

$$C_{F_\lambda} = \begin{cases} [\perp], & \lambda < -1, \\ [\Leftrightarrow], & -1 < \lambda < 1, \\ [\top], & \lambda > 1. \end{cases}$$

This gives a simple nonlinear operator trajectory

$$\perp \longrightarrow \Leftrightarrow \longrightarrow \top.$$

B.2 A two-parameter affine family

Let

$$F_{a,b,c}(x,y) = ax + by + c$$

and fix unit input magnitudes. The switching hyperplanes in (a,b,c) -space are

$$a + b + c = 0, \quad a - b + c = 0, \quad -a + b + c = 0, \quad -a - b + c = 0.$$

Their chamber sign vectors are exactly the truth tables of the corresponding linear threshold functions. This example is intentionally classical: it makes direct contact with the hyperplane-arrangement interpretation of threshold-function parameter space [7, 9] and provides a calibration point for the more general continuous polarity construction.

C Coordinate division and the rigidity mechanism

The proof of Theorem 6.8 uses only two local-to-global mechanisms. First, if a function vanishes on $x_i = 0$, coordinate division extracts a factor x_i . In the real-analytic case this is the analytic Hadamard lemma. In standard derivation-stable Denjoy–Carleman classes the same conclusion follows from the integral formula

$$G(x) = \int_0^1 \partial_i F(x_1, \dots, tx_i, \dots, x_n) dt, \quad F = x_i G,$$

with the defining derivative estimates inherited by G . Second, quasianalyticity prevents this division from continuing to infinite order for a nonzero function and prevents the terminal hyperplane restriction from vanishing on one open sign sector without vanishing identically. These two facts are precisely what convert local vanishing order into a globally uniform crossing parity.

For real-analytic functions this specializes to the familiar factorization

$$F = x_i^m G$$

with G analytic and $G|_{x_i=0}$ not identically zero. Standard analytic references include Krantz and Parks [12]; quasianalytic Denjoy–Carleman division and resolution properties are developed by Bierstone and Milman [13].

D Suggested computational verification suite

A small independent checker accompanying a later revision should verify at least the following finite statements:

1. all sixteen rows of Table 2 on the four sign assignments;
2. the composition theorem for a library of binary realizations;
3. the addition chamber split at $a = b$;

4. the five-stage affine-bias sequence of Section 9;
5. the shifted-product sequence $\perp \rightarrow \Leftrightarrow \rightarrow \top$;
6. enumeration of the eight binary affine-parity functions predicted by Corollary 6.16.

These checks would be regression tests for conventions and examples, not substitutes for the arbitrary-arity proofs.

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